GGG-P-831e
January 27, 1966

SUPERSEDING
Int. Fed. Spec. GGG-P-00831d (GSA-FSS)
August 3, 1964 and
Fed. Spec. GGG-P-831c
April 1, 1963
(See 6.3)

FEDERAL SPECIFICATION

PUNCHES; CENTER, COOPERS', DRIVE PIN, ALIGNING, AND PRICK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand punches commonly used by coopers, mechanics, and metal workers.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* Punches shall be of the following types, classes, and styles as specified (see 6.1):

Type II—Center.
  Class A—Solid.
  Class B—Automatic, adjustable.
    Style 1—With plastic knob.
    Style 2—Without plastic knob.
Type IV—Coopers'.
Type VIII—Drive pin.
  Class A—Standard point length.
    Style 1—4-inch overall length.
    Style 2—Variable overall length.
  Class B—Extra long point length.
  Class C—Solid (starting).
Type IX—Aligning
Type X—Prick.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20204.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures

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and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publication.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards (NBS) Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of punches which are otherwise in accordance with this specification.

3.2 **Material.** Punches, except as specified herein for type II, class B, styles 1 and 2, shall be of high-grade tool or alloy steel, properly hardened to produce punches having maximum resistance to breakage or deformation and complying with the requirements specified herein for each type, class, and style. All material shall be of uniform quality, free from all defects or imperfections that may adversely affect their serviceability and durability. Where tables list body size cross section as minimum, stock sizes that are undersize within commercial tolerances for bar stock are acceptable.

3.3 **Screw threads.** Screw threads of type II, class B punches shall be in accordance with NBS Handbook H28.

3.4 **Interchangeable parts (type II, class B).** All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be interchangeable, replaced, or adjusted without requiring modification.

3.5 **Finish.** Punches shall be free from oxide scale, rust, fins, sharp edges or corners (except cutting edges), pits, burrs, cracks, and other detrimental defects. Except for knurled surfaces, machined surfaces shall have a finish of 63 roughness height rating (r.h.r.) for types II and X punches. Standard commercial finish will be required for the other punches.

3.6 **Marking.** Punches shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.7 **Sets.** When sets of punches are specified (see 6.1), all punches contained in the set shall conform to the requirements of their respective type, class, and style covered by this specification.

3.7.1 *Containers for sets.* Punches, when ordered in encased sets, shall be furnished in durable wooden boxes, fabric wraps, rolls, or plastic containers as specified (see 6.1). The containers shall be provided with a compartment for each individual punch in order to facilitate selection, prevent damage to the punches, and provide a method of storage while the punches are not in use. All containers shall be marked plainly and legibly with the respective set designation.

3.7.1.1 *Wooden boxes.* The box and cover shall be of suitable hardwood such as maple, ash, hickory, or birch. Unless otherwise specified (see 6.1), the box shall have a natural finish.

3.7.1.2 *Fabric wraps or rolls.* Fabric wraps or rolls shall be of good quality, durable material and shall be provided with durable ties.

3.7.1.3 *Plastic containers.* Plastic containers shall be of a good quality plastic or plas-

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tic-coated cloth. The containers shall be provided with snap fasteners or durable ties. Plastic containers shall be capable of withstanding the test specified in 4.4.10.

3.8 **Type II, center.** Type II center punches shall be suitable for making distinct impressions on steel, brass, aluminum, and wood. Type II punches shall be capable of withstanding tests specified in 4.4.1 and 4.4.2, as applicable.

3.8.1 *Class A, solid.* Class A punches shall have the point smoothly machined or ground to the included angle specified in table I, and the tapered portion smoothly machined or ground concentric with the axis of the punch showing a runout of not more than 0.020 inch total indicator reading. The striking end of the punch shall be uniformly chamfered or rounded, and smoothly finished. The chamfering or rounding shall result in a striking-end dimension that is at least 10 percent less than the diameter, or the dimension across flats, of the adjacent portion. The face of the striking end shall be at right angles to the longitudinal axis of the punch within 2°. Class A punches shall possess a Rockwell C hardness of 59 to 63 within a minimum of 1/2 inch from the point end (see 4.4.9).

3.8.1.1 *Body shape.* The body of the punches shall be either circular, square, hexagonal, or octagonal in cross section. The punch may have a machined, or upset head that will minimize mushrooming.

3.8.1.1.1 *Circular body.* Circular body punches shall be suitably knurled and shall be provided with either a square or round striking head.

3.8.1.1.2 *Square body.* Square body punches shall have the corners suitably rounded. The square body punches shall be provided with either a round or square striking head.

3.8.1.2 Class A punches shall conform to table I and be similar to figure 1.

TABLE I. *Type II, class A, solid center punch*

Size number	Included angle of point ± 5°	Diameter at top of point (minimum) A	Body size section (minimum)	Overall length (minimum)
	Degrees	*Inch*	*Inch*	*Inches*
0	90	1/16	3/16	3-1/8
2	90	5/64	1/4	3-1/4
4	90	1/8	5/16	3-3/4
5	90	5/32	3/8	4
7	90	7/32	1/2	4
10	90	3/8	5/8	6



OPTIONAL HEAD DESIGN



FIGURE 1. Type II, class A, solid center punch.

3.8.2 *Class B, automatic, adjustable.* Class B punches shall consist essentially of a high-grade tool-steel punch point, a brass, steel or aluminum-alloy body and screwcap, and an internal mechanism which automatically strikes a blow of a required force by a movement of the body of the punch in the direction of the point. The working parts shall be suitably hardened to minimize wear, distortion or fracture, and shall be accessible for replacement. Brass bodies and brass screwcaps shall be suitably plated to resist corrosion. The point end of class B punches shall have a Rockwell C hardness of from 59 to 63 with the exception of the threaded portion. The bodies and screwcaps, when made of aluminum alloy, shall have a Brinell hardness of from 117 to 125 (see 4.4.9).

3.8.2.1 *Punch point.* The punch point shall have a center point smoothly machined or ground concentric with the punch-point body and, when assembled, shall show a runout of not more than 0.020-inch total indicator reading. The punch point shall be removable for resharpening, and shall be easily replaced. The point shall have an included angle of 80° ± 5°.

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3.8.2.2 *Body.* The outer diameter of the body shall be suitably knurled. The inner portion of the body shall be designed for proper housing of the automatic mechanism and punch point.

3.8.2.3 *Screwcap.* The outer diameter of the screwcap shall be suitably knurled. The screwcap shall be adjustable on the body, working in conjunction with a spring which shall regulate the force of the stroke.

3.8.2.4 *Style 1, with plastic knob.* Style 1 center punches shall be provided with a plastic knob securely attached to the adjusting screwcap in such a manner that it will not become loose or detached under normal service conditions. The plastic knob shall be round and fluted or otherwise suitably designed to promote hand comfort. Plastic knobs with a cellulose-nitrate base shall not be used. Plastic knobs shall withstand the tests specified in 4.4.7. Style 1 center punch shall conform to table II and shall be similar to figure 2.

TABLE II. *Type II, class B, style 1, automatic, adjustable center punch with plastic knob*

Size	Diameter of body + 1/16 − 1/64	Length overall extended[1] (maximum)	Length overall closed[1] (minimum)
	Inch	*Inches*	*Inches*
1	1/2	4-1/2	4-1/4

[1] Extended and closed dimensions are obtained by tightening and loosening the screwcap to adjust the force of the stroke.

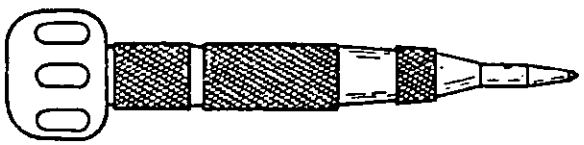


FIGURE 2. Type II, class B, style 1, automatic, adjustable center punch with plastic knob.

3.8.2.5 *Style 2, without plastic knob.* Style 2 center punch shall be furnished without a plastic knob, shall conform to table III, and be similar to figure 3.

TABLE III. *Type II, class B, style 2, automatic, adjustable center punch without plastic knob*

Size number	Diameter of body + 1/16 − 1/64	Overall length closed (minimum)	Pressure required to actuate mechanism[1]	
			Minimum	Maximum
	Inch	*Inches*	*Pounds*	*Pounds*
1	3/8	3-1/8	5	15
2	1/2	4-1/8	10	25
3	5/8	5	40	55

[1] With the screwcap in closed position, the required load to actuate the mechanism shall fall within the specified range (when tested on a platform scale).



FIGURE 3. Type II, class B, style 2, automatic, adjustable center punch without plastic knob.

3.9 **Type IV, coopers'.** Type IV punches shall be approximately 4-1/2 inches in length, provided with a 3/16-inch diameter cup point. The striking end shall be uniformly chamfered, and shall be smoothly finished. The chamfering shall result in a striking-end diameter that is at least 10 percent less than the diameter, or the dimension across flats, of the adjacent portion. Type IV punches shall show a Rockwell C hardness of from 50 to 59, or a hardness of 85.5 to 89.5 on the Rockwell superficial 15 *N* scale (see 4.4.9). The hardness reading shall be taken within a minimum of 1/2 inch from the cutting end of the punch. Type IV punches shall be capable of withstanding the test specified in 4.4.3.

3.9.1 *Body.* The body diameter shall be approximately 1/2 inch either circular, hexagonal, or octagonal in cross section. Circular body punches shall be suitably knurled.

3.9.2 *Tapered point.* The tapered point shall be approximately 2 inches in length, smoothly finished, and concentric with the axis of the body showing a runout of not more than 0.020 inch total indicator reading. The cutting edge of the tapered point

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shall not become easily dulled when subjected to the test specified in 4.4.3.

3.9.3 Type IV punches shall be similar to figure 4.

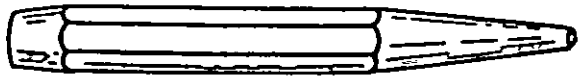


FIGURE 4. Type IV, coopers' punch.

3.10 Type VIII, drive pin. Type VIII punches shall be suitable for use in driving pins or bolts from holes having a force or drive fit. The striking-end face of the punch shall be uniformly chamfered and also smoothly machined or ground at right angles to the longitudinal axis of the punch within 2°. The chamfering shall result in a striking-end dimension that is at least 10 percent less than the diameter, or the dimension across flats, of the adjacent portion. The entire length of the point of the punches shall be smoothly machined or ground concentric with the axis of the body showing a runout of not more than 0.020-inch total indicator reading. Type VIII punches shall be capable of withstanding the test specified in 4.4.4 or 4.4.5, as applicable. Type VIII, classes A, B, and C punches shall have a Rockwell C hardness of 59 to 63 throughout their point length.

3.10.1 *Body shape.* The body of the punches shall be either circular, hexagonal, octagonal, or square in cross section. The punch may have an upset head that will minimize mushrooming (see optional head design fig. 5 and 6).

3.10.1.1 *Circular body.* Circular-body punches shall be suitably knurled in such a manner as to facilitate a firm grip under normal service conditions.

3.10.2 *Class A, standard-point length.*

3.10.2.1 *Style 1, 4-inch overall length.* Class A, style 1 punches shall be furnished in normal sets, special sets, or individual punches as specified (see 6.1). Normal sets shall consist of one each of sizes 1 through 8. Special sets shall consist of any combination of sizes required. Class A, style 1 punches shall conform to table IV and be similar to figure 5a or 5b.

TABLE IV. *Type VIII, class A, style 1, drive pin punch, standard point length, 4-inch overall length*

Size number	Diameter of point + 0.000 − 0.015 A	Point length (minimum) B	Body size cross section (minimum) C	Length, overall ± 1/8 D
	Inch	*Inch*	*Inch*	*Inches*
1	1/16	1/2	1/4	4
2	3/32	11/16	1/4	4
3	1/8	3/4	9/32	4
4	5/32	13/16	9/32	4
5	3/16	15/16	5/16	4
6	7/32	1	5/16	4
7	1/4	1	3/8	4
8	5/16	1	3/8	4
9	3/8	1	1/2	4

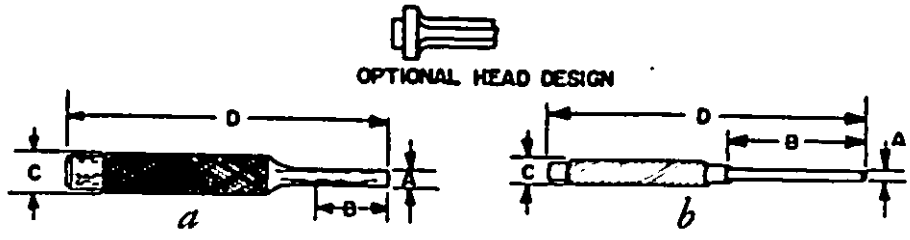


FIGURE 5. Type VIII, class A, styles 1 and 2 and class B drive-pin punches.

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3.10.2.2 *Style 2, variable-overall length.* Class A, style 2 punches shall conform to table V and be similar to figure 5a or 5b.

TABLE V. *Type VIII, class A, style 2, drive pin punch, standard point length, variable overall length*

Size number	Diameter of point + 0.000 − 0.015 A	Point length (minimum) B	Body size cross section (minimum) C	Length, overall (minimum) D
	Inch	*Inches*	*Inch*	*Inches*
1	1/16	1/2	1/4	3
2	3/32	11/16	1/4	4
3	1/8	3/4	5/16	4-1/2
4	5/32	13/16	5/16	5
5	3/16	1	5/16	5
10	3/16	1-1/2	5/16	6
6	7/32	1	3/8	5
7	1/4	1	3/8	5-1/2
8	5/16	1	7/16	6
9	3/8	1	1/2	6

3.10.3 *Class B, extra-long point length.* Class B punch shall conform to table VI and be similar to figure 5a or 5b.

TABLE VI. *Type VIII, class B, drive pin punch, extra-long point length*

Size number	Diameter of point + 0.000 − 0.015	Point length (minimum)	Body size, cross section (minimum) C	Length overall (minimum) D
	Inch	*Inches*	*Inch*	*Inches*
0	1/16	1-1/2	1/4	6
1	1/8	3-1/2	3/8	8
1a	5/32	2	3/8	8
1	1/8	3-1/2	3/8	8
2	3/16	3-1/2	1/2	8
3	1/4	3-1/2	1/2	8
4	5/16	3-1/2	1/2	8
5	3/8	3-1/2	9/16	8
6	1/2	3-1/2	5/8	8

3.10.4 *Class C, solid (starting).* Class C punches shall conform to table VII and be similar to figure 6.

TABLE VII. *Type VIII, class C, drive pin punch, solid (starting)*

Size number	Diameter of point + 0.000 − 0.015 A	Taper length B		Body size cross section C (minimum)	Length overall D (minimum)
		Minimum	Maximum		
	Inch	*Inches*	*Inches*	*Inch*	*Inches*
1	1/16	1-1/2	2-1/2	5/16	4
1a	3/32	1-1/2	2-1/2	5/16	4
2	1/8	1-1/2	2-3/4	5/16	4-1/2
3	3/16	1-3/4	3	3/8	5
4	1/4	2	3-1/2	1/2	5-1/2
5	5/16	2	3-1/2	1/2	5-3/4
6	3/8	2	4	5/8	6
8	1/2	2-1/2	5	5/8	6-1/2

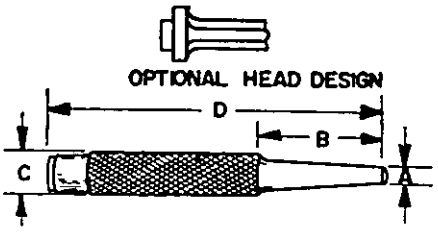


FIGURE 6. Type VIII, class C, drive pin punch, solid (starting).

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TABLE VIII. *Type IX, aligning punch*

Size number	Diameter of point + 0.000 − 0.015 A	Taper length B		Body size cross section C (minimum)	Length overall D (minimum)	Bending moment
		Minimum	Maximum			
	Inch	*Inches*	*Inches*	*Inch*	*Inches*	*Inch-pounds*
1	3/32	3-1/2	5	1/4	8	30
2	1/8	3-1/2	5	5/16	8	50
3	5/32	4	6-1/2	3/8	8	70
4	5/32	6	8-1/2	1/2	12	70
5	3/16	4	6-1/2	3/8	8	100
6	3/16	6	8-1/2	9/16	12	100
7	7/32	4	6-1/2	7/16	8	130
8	1/4	4	6-1/2	7/16	8-1/2	300
9	1/4	7	8-1/2	5/8	12	300
10	5/16	4	6-1/2	1/2	8-1/2	500
11	5/16	7	8-1/2	5/8	12	500
12	3/8	4	6-1/2	9/16	9	800
13	3/8	7	8-1/2	3/4	12	800
14	1/2	10	12	3/4	18	1,000

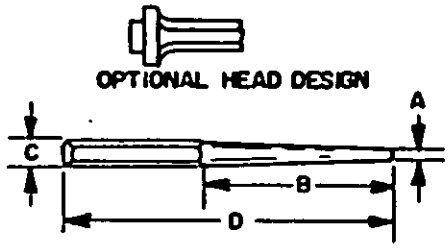


FIGURE 7. Type IX, aligning punch.

3.11 Type IX, aligning. Type IX punches shall be suitable for lining up holes in plates, frames, and housings by either hammering or prying. The striking-end face of the punch shall be uniformly chamfered and smoothly finished at right angles to the longitudinal axis of the body. The chamfering shall result in a striking-end dimension that is at least 10 percent less than the diameter or the dimension across flats of the adjacent portion. The taper length shall be smoothly finished and concentric with the axis of the body showing a runout of not more than 0.060-inch total indicator reading. Type IX punches shall be capable of withstanding the bending moment test specified in 4.4.8. Type IX punches shall have a Rockwell C hardness of 59 to 63 throughout a minimum of 75 percent of the taper length from the point of the taper.

3.11.1 *Body shape.* The body of the punches shall be either circular, hexagonal, octagonal, or square in cross section. The punch may have an upset head that will minimize mushrooming.

3.11.1.1 *Circular body.* Circular-body punches shall be suitably knurled in such a manner as to facilitate a firm grip under normal service conditions.

3.11.1.2 Type IX punches shall conform to table VIII and be similar to figure 7.

3.12 Type X, prick. Type X punch points shall be a 50° ± 5° included angle. The tapered portion shall be machined or ground concentric with the axis of the punch showing a runout of not more than 0.020-inch total indicator reading. The striking end of the punch shall be uniformly chamfered, rounded, or upset to minimize mushrooming and shall be smoothly finished. The face of the striking end shall be at right angles to the longitudinal axis of the punch within 2°. The point end shall have a Rockwell C hardness of 59 to 63 within a minimum of 1/2 inch from the point end. Punches shall conform to table IX and be similar to figure 8. Type X punches shall be able to withstand the test specified in 4.4.6.

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TABLE IX. *Type X, prick punch*

Size number	Body size, cross section (minimum)	Length overall (minimum)
	Inch	*Inches*
1	1/4	4
1a	5/16	4
2	3/8	4
3	1/2	5

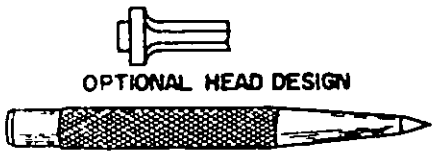


FIGURE 8. Type X, prick punch.

3.13 Workmanship. The workmanship shall be first class throughout.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table X.

4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, construction, workmanship, and marking.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.3.3 *Examination of containers and sets.* Each sample unit shall be examined for any nonconformance with assembly of sets, including the boxes or rolls (see 3.7 through 3.7.1.3).

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.10.

4.4.1 *Performance test for type II center punches.* Repeated impressions shall be made in steel plate (free of scale) at least

TABLE X. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	2.5	Defects per hundred units	4.3.1
Dimensional examination	do.	II	2.5	do.	4.3.2
Examination of containers and sets*	do.	S-2	6.5	do.	4.3.3
Testing	do.	S-4	4.0	Percent defective	4.4.1 thru 4.4.10
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.5

*Note: *Punch set.* Each item in the set, including container, shall be considered as an individual item and inspected accordingly.

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1/4 inch thick and having a hardness between 35 and 40 on the Rockwell C scale. A machinists' hammer (or any mechanical device which in the opinion of the inspector will give comparable and satisfactory results) shall be used in making these impressions. Solid punches with point diameters of 3/32 inch and under, and all points from automatic adjustable center punches shall be required to punch at least 1000 impressions 0.03 inch (minimum) in depth. Points of automatic center punches shall be removed and secured in a separate steel shank of suitable size for purposes of this test. Solid punches with point diameters larger than 3/32 inch up to and including 3/16 inch shall be required to punch at least 1000 impressions 0.045 inch (minimum) in depth. Solid punches with point diameters larger than 3/16 inch shall be required to punch at least 1000 impressions 0.07 inch (minimum) in depth. Punches shall perform without indication of failure. Failure shall be understood to have occurred should the sample chip, spall, or mushroom, the body become bent, or the point become fractured, or blunted. If any punch in the sample fails to conform to the performance test, a second sample shall be taken and submitted to the same test. If any punch in this second sample fails, the entire lot shall be rejected.

4.4.2 *Functional test for type II, center punch, class B, automatic, adjustable, styles* 1 *and* 2. Each sample punch shall be cycled 100 times to determine whether the automatic device functions properly, and whether the adjustment regulates pressure properly.

4.4.3 *Performance test for type IV, coopers'*. The sample punches shall be tested by punching at least 100 holes in soft low carbon sheet steel of approximately 0.026 inch in thickness. The cutting edge shall show ease in cutting and demonstrate the ability to hold a keen edge under normal service conditions. Attention shall be given to each sample punch as to whether the striking end, shank, body, and taper offer proper resistance to breakage, deformation, spalling, or cracking.

4.4.4 *Performance test for type VIII, drive pin punch, class A and class B, standard and extra-long point length.* The sample punches shall be tested by removing 12 or more steel pins from holes into which the pins have been inserted with a forced or drive fit. The plate shall be approximately equal in thickness to the length of the point but not over 1 inch thick. The length of the pins shall be not less than 65 percent nor more than 85 percent of the plate thickness to allow the punch point to enter the drive pin holes. Each sample punch shall be capable of removing the pins from the plate and shall demonstrate the ability to resist breakage, deformation, spalling, or cracking.

4.4.5 *Performance test for type VIII, drive pin punch, class C, solid (starting).* The sample punches shall be tested by starting 12 or more steel drive pins from holes into which the drive pins have been inserted with a forced or drive fit. A 1 inch thick steel plate or plates, fitted with the required number of pins for each sample punch, shall be used for the performance test. The diameter of the drive pins shall be 1/16 inch larger than the point diameter of the sample punch. The pins shall not exceed 1 inch in length. Each sample punch shall be capable of starting and moving the pins approximately one-quarter inch and shall demonstrate the ability to resist any breakage, deformation, spalling, or cracking.

4.4.6 *Performance test for type X, prick punch.* Each sample shall be tested under service conditions on steel plate free of scale at least 1/8 inch in thickness having a hardness between 20 and 25 on the Rockwell C scale. A machinists' hammer shall be used in the test and at least 100 impressions 0.03 inch (minimum) in depth shall be made. Attention shall be given to each sample punch as to whether the striking end and point offer proper resistance to breakage, deformation, dulling, spalling, or cracking.

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4.4.7 *Plastic knob test, type II, center, class B, automatic, adjustable style* 1, *plastic knob*. The punches shall be subjected to the tests specified in 4.4.7.1, 4.4.7.2, and 4.4.7.3.

4.4.7.1 *Heat-distortion test.* The knob of the punch shall be completely immersed in boiling water at 212°F. for a period of 2 minutes. At the end of this period, the punch shall be removed from the water and the knob shall show no significant signs of distortion or blistering.

4.4.7.2 *Impact test at room temperature.* Each sample punch shall be held knob downward and dropped on a concrete floor from a height of 10 feet. After impact there shall be no breakage, loosening, buckling, spalling, or cracking of the knob.

4.4.7.3 *Flammability test.* The body of the punch shall be clamped in a support with the knob extended outward in a horizontal position. There shall be clamped in a horizontal position, a 5-inch square piece of 10-mesh Bunsen burner gauze, one-quarter inch below the bottom of the knob and one-half inch back from the butt end of the knob. A Bunsen burner with an efficient flame 1/2 to 3/4 inch in height shall be adjusted so that the flame tip will contact the specimen when placed under the free end of the knob. At the end of 30 seconds, the flame shall be removed and the sample allowed to burn. The time required for the flame to travel from the butt end to the screw cap end of the knob after the removal of the Bunsen burner shall determine the rate of travel of the flame. In case the knob does not continue to burn after the first ignition, the Bunsen burner shall be replaced under the free end of the knob for a period of 30 seconds immediately following the extinction of the flame on the handle. If the knob stops burning before the flame has reached the screwcap end of the knob after the second ignition, the sample shall be considered self-extinguishing. When tested as specified herein, the plastic knob shall not burn at a rate exceeding 2 inches per minute.

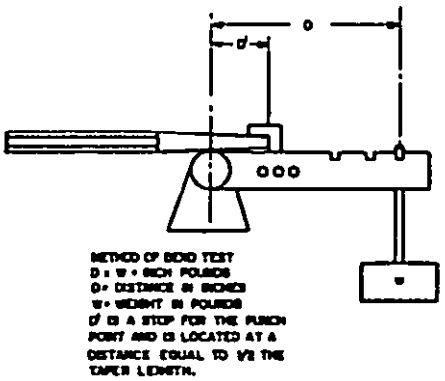
4.4.8 *Bending moment test for type IX, aligning punch.* Each sample punch shall be tested for the bending moment as follows. The test shall be conducted in a manner similar to that shown on figure 9 to determine conformance with the requirements shown in table VIII. In this test the force shall be applied near the middle of the body, the force to act substantially at right angles to the axis of the tool. The load on the point end shall be supported not more than 1/8 inch from the end of the point of the taper. A spring balance or other suitable load measuring device may be used in lieu of a dead weight incident to applying the inch pounds bending test. When tested in this manner, the punch shall not fracture nor show any visible permanent set.

4.4.9 *Hardness test.* The sample punches shall be tested in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine compliance with the hardness requirement of the respective types.

4.4.10 *Flammability test for plastic wrap or roll.* The plastic wrap or roll shall be rolled into a tight roll and clamped in a support, outward in a horizontal position. There shall be clamped in a horizontal position a 5-inch square piece of 10-mesh Bunsen burner gauze, one-quarter inch below the bottom of the roll and one-half inch back from the free end of the roll. A Bunsen burner with an efficient flame 1/2 to 3/4 inch in height shall be adjusted so that the flame tip will contact the specimen when placed under the free end of the roll. At the end of 30 seconds the flame shall be removed and the sample allowed to burn. The time required for the flame to travel from the end of the roll to the clamping device after the removal of the Bunsen burner shall determine the rate of travel of the flame. In case the roll does not continue to burn after the first ignition, the Bunsen burner shall be replaced under the free end of the roll for a period of 30 seconds immediately following the extinction of the flame on the roll. If the roll stops burning before the flame has reached the clamp-

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ing device after the second ignition, the sample shall be considered self-extinguishing. When tested as specified, the plastic roll shall not burn at a rate exceeding 4 inches per minute.



NOTE: The above method is not intended to restrict the manner in which the required tests shall be taken.

FIGURE 9. Bending moment test.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table XI. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE XI. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.1), the punches shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation, and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.1).

5.1.1 As part of the requirements, section 5 of this specification, table XII is considered a part of table I of MIL-H-15424.

TABLE XII. *(Addition to table I of MIL-H-15424)*

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Punches; center:						
Solid	P-1	I	1	10	5	250
Automatic	P-8, P-9	IC-1	1	12	30	360
Coopers	P-1	I	12	12	4	576
Drive pin:						
Through 1/4 in. dia	P-1	I	12	12	4	576
5/16 through 1/2 in. dia.	P-1	I	12	6	4	288
5/8 and over	P-1	I	6		12	72
Aligning	P-1	I	12	12	4	576
Prick	P-1	I	12	12	4	576
Set:						
Punches	None	IC-1	—	—	36	

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5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and style (if applicable) of punches required (see 1.2).
(*c*) Size number of punches required (see applicable tables).
(*d*) Whether punches in sets are required (see 3.7).
(*e*) Whether sets should be furnished in wooden boxes, fabric wraps or rolls or plastic containers (see 3.7.1).
(*f*) If special finish is required for box (see 3.7.1.1).
(*g*) Whether normal set, special set as required, or individual punches of type VIII, class A, style 1 are required (see 3.10.2.1).
(*h*) Level of preservation, packaging, and packing required (see 5.1).

6.2 **Transportation descriptions.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name. Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, not otherwise indexed. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.3 **Supersession data.** Type VI (cutting, drive) punch formerly included in Federal Specification GGG-P-831c is now covered by Federal Specification GGG-P-833a, Punches, Cutting.

DOD coordination has been waived.

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